

Appendix A：穩健性檢驗：OLS 與負二項迴歸模型比較

為檢驗研究結果是否受到模型設定方式影響，本研究進一步以適用於計數資料的 Negative Binomial Regression 重新估計模型。結果顯示，降雨、溫度、溫度平方項、濕度與週末效果之估計方向及統計顯著性均與 OLS 模型一致。其中，降雨仍為影響 YouBike 需求最重要的負向因素，而溫度則呈現倒 U 型關係。整體而言，主要研究發現不因模型選擇而改變，顯示本文結果具有良好的穩健性（robustness）。

表 A1 OLS 與負二項迴歸模型比較

	OLS	Negative Binomial
Rain Dummy	-22.835*** (0.350)	-0.521*** (0.005)
Temperature	15.184*** (0.584)	0.290*** (0.008)
Temperature ²	-0.236*** (0.010)	-0.004*** (0.000)
Humidity	-0.313*** (0.015)	-0.004*** (0.000)
Weekend	-14.071*** (0.180)	-0.172*** (0.002)

Appendix B：騎乘距離分析與借還車分析

圖 B1 顯示捷運周邊 YouBike 騎乘距離呈現明顯右偏（right-skewed）分布，大部分騎乘集中於短距離區間。樣本騎乘距離中位數約為 0.73 公里，表示有一半以上的騎乘旅次距離不到 1 公里；此外，90% 的旅次距離低於 1.87 公里。雖然少數旅次可達 4 至 5 公里以上，但其比例相當有限。此結果顯示，本研究所識別之捷運周邊 YouBike 使用行為確實以短程移動為主，符合接駁運輸的特徵。換言之，多數使用者是利用 YouBike 完成捷運站與目的地之間的短距離移動，而非將 YouBike 作為長距離主要運輸工具。

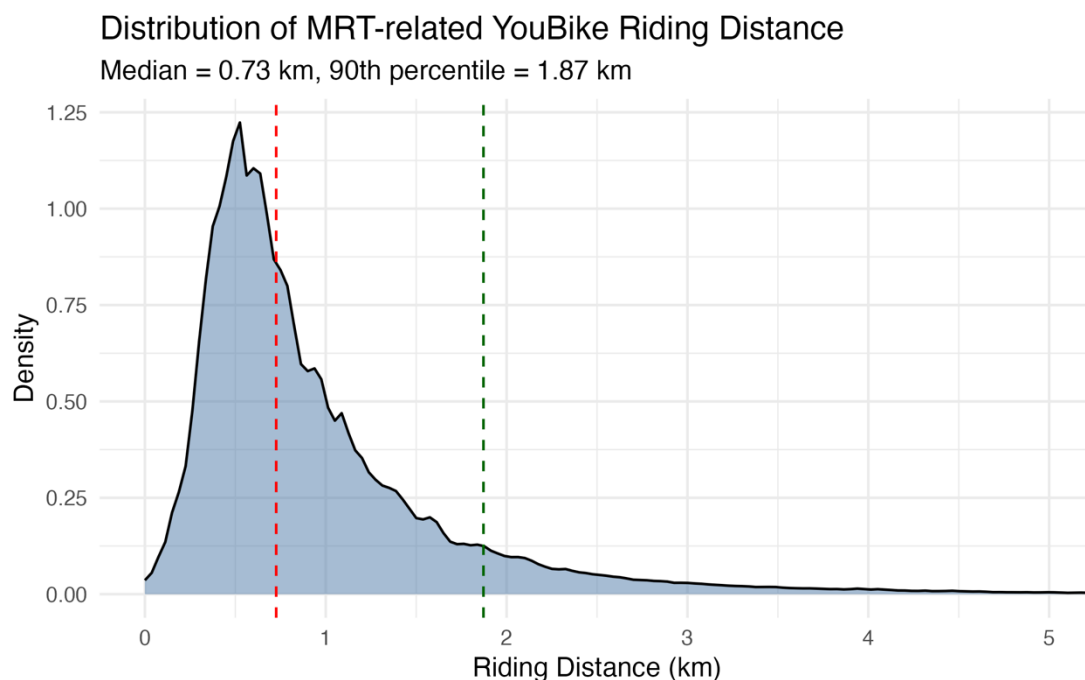


圖 B1 捷運周邊 Youbike 騎乘距離分布

圖 B2 比較各時段之借車（Borrow）與還車（Return）需求，兩者皆呈現明顯的雙峰型態。第一波高峰出現在上午 7 - 8 時，第二波高峰則出現在下午 17 - 18 時，與一般通勤族上下班時段高度一致。

此外，借車與還車曲線整體走勢相當接近，顯示捷運周邊 YouBike 系統大致維持供需平衡。然而，在晚間尖峰時段（18 - 21 時）借車需求略高於還車需求，而上午通勤時段（6 - 7 時）則觀察到還車需求略高於借車需求，反映部分站點可能具有單向通勤流動特徵。

整體而言，此圖進一步支持捷運周邊 YouBike 主要扮演通勤接駁工具的角色，其需求型態與都市居民的日常通勤作息高度一致。

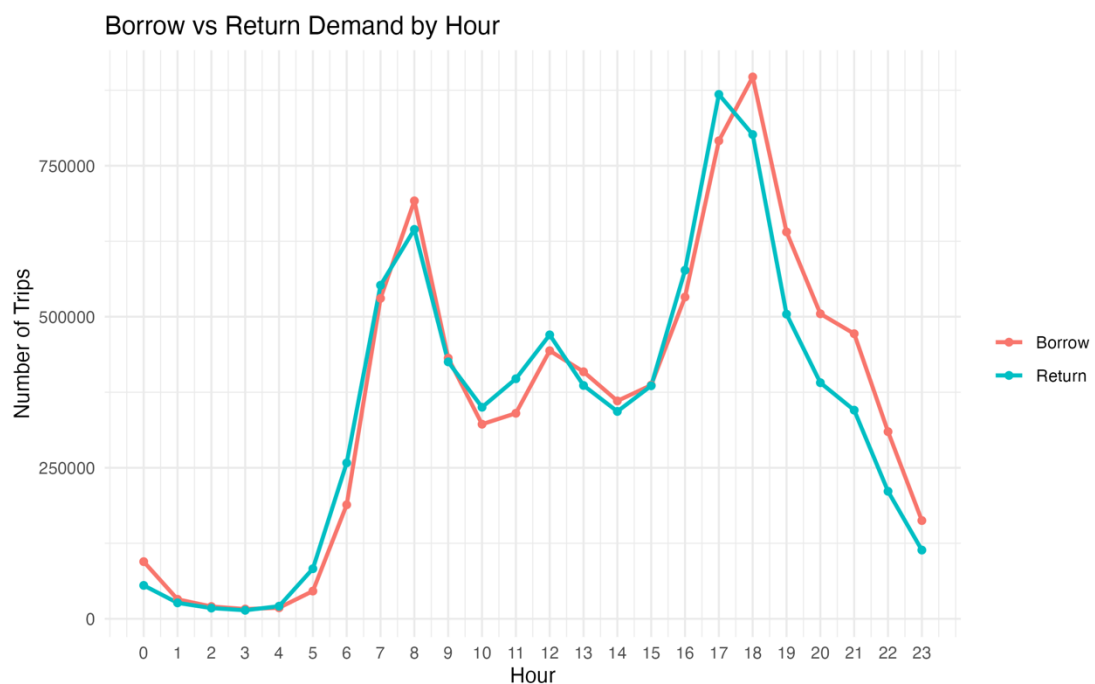


圖 B 2 Youbike 借還車需求之小時分布

Appendix C : R script

C.1 : Weather data

```
library(jsonlite)
library(dplyr)
library(sf)
library(stringr)
library(tidyr)
library(lubridate)
library(purrr)

#讀天氣資料
temp_files <- list.files(
  "Data/weather/temp",
  pattern = "AirTemperature-hour.csv",
  full.names = TRUE
)

read_temp_file <- function(file_path) {
  temp_raw <- read.csv(file_path)
  file_name <- basename(file_path)
  file_parts <- str_split(
    file_name,
    "-",
    simplify = TRUE
  )
  station_id <- file_parts[1]
  year_month <- paste0(
    file_parts[2],
    "-",
    file_parts[3]
  )
  message(
    "reading: ", file_name,
    " | station_id = ", station_id,
    " | year_month = ", year_month
  )
}
```

```

temp_long <- temp_raw %>%
  mutate(
    across(
      starts_with("X"),
      as.character
    ),
    日.時 = as.character(日.時),
    day = str_extract(日.時, "\\d+"),
    day = as.numeric(day)
  ) %>%
  filter(
    !is.na(day),
    day >= 1,
    day <= 31
  ) %>%
  pivot_longer(
    cols = starts_with("X"),
    names_to = "hour",
    values_to = "temperature"
  ) %>%
  mutate(
    hour = as.numeric(
      gsub("X", "", hour)
    ),
    date_str = paste0(
      year_month,
      "-",
      sprintf("%02d", day)
    ),
    date = suppressWarnings(
      ymd(date_str)
    ),
    temperature = str_trim(temperature),
    temperature = case_when(
      temperature %in% c(
        "&",
        "X",
        "/"
      )
    )
  )

```

```

      "-",
      "",
      "NA"
    ) ~ NA_character_,
    temperature == "T" ~ "0",
    TRUE ~ temperature
  ),
  temperature = as.numeric(temperature),
  weather_station_id = station_id
) %>%
filter(!is.na(date)) %>%
select(
  weather_station_id,
  date,
  hour,
  temperature
)
return(temp_long)
}

```

```

temp_panel <- temp_files %>%
  map_dfr(read_temp_file)
saveRDS(temp_panel, "Output/temp_panel.rds")

```

#讀雨量資料

```

rain_files <- list.files(
  "Data/weather/rain",
  pattern = "Precipitation-hour.csv",
  full.names = TRUE
)

```

```

read_rain_file <- function(file_path) {
  rain_raw <- read.csv(file_path)
  file_name <- basename(file_path)
  file_parts <- str_split(
    file_name,
    "-",

```

```

    simplify = TRUE
  )
  station_id <- file_parts[1]
  year_month <- paste0(
    file_parts[2],
    "-",
    file_parts[3]
  )
  message(
    "reading: ", file_name,
    " | station_id = ", station_id,
    " | year_month = ", year_month
  )
  rain_long <- rain_raw %>%
    mutate(
      across(
        starts_with("X"),
        as.character
      ),
      日.時 = as.character(日.時),
      day = str_extract(日.時, "\\d+"),
      day = as.numeric(day)
    ) %>%
    filter(
      !is.na(day),
      day >= 1,
      day <= 31
    ) %>%
    pivot_longer(
      cols = starts_with("X"),
      names_to = "hour",
      values_to = "rain"
    ) %>%
    mutate(
      hour = as.numeric(
        gsub("X", "", hour)
      ),
      date_str = paste0(

```

```

    year_month,
    "-",
    sprintf("%02d", day)
  ),
  date = suppressWarnings(
    ymd(date_str)
  ),
  rain = str_trim(rain),
  rain = case_when(
    rain %in% c(
      "&",
      "X",
      "/",
      "-",
      "",
      "NA"
    ) ~ NA_character_,
    rain == "T" ~ "0",
    TRUE ~ rain
  ),
  rain = as.numeric(rain),
  weather_station_id = station_id
) %>%
# 去掉日期
filter(!is.na(date)) %>%
select(
  weather_station_id,
  date,
  hour,
  rain
)

return(rain_long)
}

library(purrr)

rain_panel <- rain_files %>%

```

```
map_dfr(read_rain_file)
saveRDS(rain_panel, "Output/rain_panel.rds")
```

#讀濕度資料

```
humidity_files <- list.files(
  "Data/weather/humidity",
  pattern = "RelativeHumidity-hour",
  full.names = TRUE
)

read_humidity_file <- function(file_path) {
  humidity_raw <- read.csv(file_path)
  file_name <- basename(file_path)
  file_parts <- str_split(
    file_name,
    "-",
    simplify = TRUE
  )
  station_id <- file_parts[1]
  year_month <- paste0(
    file_parts[2],
    "-",
    file_parts[3]
  )
  message(
    "reading: ", file_name,
    " | station_id = ", station_id,
    " | year_month = ", year_month
  )
  humidity_long <- humidity_raw %>%
    mutate(
      across(
        starts_with("X"),
        as.character
      ),
      日.時 = as.character(日.時),
      day = str_extract(日.時, "\\d+"),
      day = as.numeric(day)
```



```

) %>%
filter(
  !is.na(day),
  day >= 1,
  day <= 31
) %>%
pivot_longer(
  cols = starts_with("X"),
  names_to = "hour",
  values_to = "humidity"
) %>%
mutate(
  hour = as.numeric(
    gsub("X", "", hour)
  ),
  date_str = paste0(
    year_month,
    "-",
    sprintf("%02d", day)
  ),
  date = suppressWarnings(
    ymd(date_str)
  ),
  humidity = str_trim(humidity),
  humidity = case_when(
    humidity %in% c(
      "&",
      "X",
      "/",
      "-",
      "",
      "NA",
      "...")
    ) ~ NA_character_,
    TRUE ~ humidity
  ),
  humidity = as.numeric(humidity),
  weather_station_id = station_id

```

```

    ) %>%
    filter(!is.na(date)) %>%
    select(
      weather_station_id,
      date,
      hour,
      humidity
    )
    return(humidity_long)
  }

humidity_panel <- humidity_files %>%
  map_dfr(read_humidity_file)

saveRDS(humidity_panel, "Output/RelativeHumidity.rds")

```

C.2 : Data-cleaning

```

library(jsonlite)
library(dplyr)
library(sf)
library(stringr)
library(tidyr)
library(lubridate)
library(geosphere)

## cleaning pipeline
#1. 讀資料
#2. 建立 YouBike ↔ MRT 對照表
#3. 建立 YouBike ↔ 氣象站對照表
#4. 讀 6 - 9 月騎乘資料
#5. 篩 MRT-substitution trips
#6. aggregate 成 MRT x date x hour
#7. 讀天氣資料
#8. merge 天氣
#9. 輸出 final data

#捷運出入口資料

```

```

mrt <- read.csv("Data/MRT.csv", fileEncoding = "Big5")

#Youbike 站點資料
url <-
"https://tcgbusfs.blob.core.windows.net/dotapp/youbike/v2
/youbike_immediate.json"
bike <- fromJSON(url)
bike_station <- bike %>% select(sno, sna, sarea,
latitude, longitude)
head(bike_station)
bike_sf <- st_as_sf(
  bike_station,
  coords = c("longitude", "latitude"),
  crs = 4326
)
bike_sf <- st_transform(bike_sf, 3826)

#配對 200 公尺內有 Youbike 站的捷運站
mrt_sf <- st_as_sf(mrt, coords = c("經度", "緯度"), crs =
4326) #WGS84 全球經緯度系統
mrt_sf <- st_transform(mrt_sf, 3826) #轉成：「台灣的公尺座標
系」TM2
mrt_buffer <- st_buffer(mrt_sf, dist = 200)
nearby <- st_join(bike_sf, mrt_buffer, join = st_within)
mrt_bike_station <- nearby %>%
  filter(!is.na(出入口名稱)) %>% distinct(sna) #選定在捷運站入
口附近兩百公尺的站點
#mrt_bike_station <- mrt_bike_station %>%
# mutate(
#   sna = gsub("YouBike2.0_", "", sna)
# )

#將 youbike 站配對到最近捷運站
mrt_bike_station_map <- nearby %>%
  filter(!is.na(出入口名稱)) %>%
  st_drop_geometry() %>%
  mutate(

```

```

    sna = gsub("YouBike2.0_", "", sna),
    mrt_station = str_extract(出入口名稱, ".*?站")
  ) %>%
  select(
    sna,
    mrt_station,
    出入口名稱
  ) %>%
  distinct()

#騎乘資料
trip_files <- list.files(
  "Data/票證資料",
  pattern = "YouBike2.0 票證刷卡資料.csv",
  full.names = TRUE
)

read_youbike_trip <- function(file_path) {
  trip <- read.csv(
    file_path,
    header = FALSE
  )
  names(trip) <- c(
    "start_time",
    "start_station",
    "end_time",
    "end_station",
    "duration",
    "bike_type",
    "date"
  )
  trip <- trip %>%
  mutate(
    start_station = trimws(start_station),
    end_station = trimws(end_station),
    start_time = ymd_hms(start_time),
    date = as.Date(start_time),
    hour = hour(start_time)
  )
}

```

```

    )
  return(trip)
}

Youbike_all <- trip_files %>%
  lapply(read_youbike_trip) %>%
  bind_rows()

Youbike_all <- Youbike_all %>%
  mutate(
    start_station = trimws(start_station),
    end_station = trimws(end_station)
  ) #讓站名字串統一

#篩選出符合騎乘起站以及終站都在捷運站附近的站點
Youbike_all <- Youbike_all %>%
  mutate(
    start_near_mrt = start_station %in%
mrt_bike_station_map$sna
  )

mrt_sub_Youbike_all <- Youbike_all %>%
  filter(
    (start_near_mrt) & (start_station != end_station)
  )

#配對 youbike 到捷運站
nearest_mrt_idx <- st_nearest_feature(bike_sf, mrt_sf)
bike_mrt_map <- bike_sf %>%
  mutate(
    nearest_mrt_exit = mrt_sf$出入口名稱[nearest_mrt_idx],
    sna = gsub("YouBike2.0_", "", sna),
    mrt_station = str_extract(nearest_mrt_exit, ".*?站")
  ) %>%
  st_drop_geometry() %>%
  select(sna, mrt_station, nearest_mrt_exit) %>%
  distinct(sna, .keep_all = TRUE)

```

```

# Borrow / Return 分開 + 騎乘距離
# 建立 YouBike 站點座標表
clean_station_name <- function(x) {
  x %>%
    str_replace("^YouBike2.0_", "") %>%
    str_replace("_\\d+$", "") %>%
    str_trim()
}

bike_coord <- bike_station %>%
  mutate(
    sna_clean = clean_station_name(sna)
  ) %>%
  group_by(sna_clean) %>%
  summarise(
    latitude = first(latitude),
    longitude = first(longitude),
    .groups = "drop"
  )

mrt_sub_Youbike_all <- mrt_sub_Youbike_all %>%
  mutate(
    start_station_clean =
      clean_station_name(start_station),
    end_station_clean = clean_station_name(end_station)
  )

# 加入起點與終點座標
mrt_sub_Youbike_all <- mrt_sub_Youbike_all %>%
  left_join(
    bike_coord %>%
      rename(
        start_lat = latitude,
        start_lng = longitude
      ),
    by = c("start_station_clean" = "sna_clean")
  ) %>%
  left_join(

```

```

bike_coord %>%
  rename(
    end_lat = latitude,
    end_lng = longitude
  ),
  by = c("end_station_clean" = "sna_clean")
)

# 計算起訖站直線距離，單位為 km
mrt_sub_Youbike_all <- mrt_sub_Youbike_all %>%
  mutate(
    ride_distance_km =
      distHaversine(
        cbind(start_lng, start_lat),
        cbind(end_lng, end_lat)
      ) / 1000,
  )

mrt_sub_Youbike_all <- mrt_sub_Youbike_all %>%
  left_join(
    bike_mrt_map %>% select(sna, mrt_station),
    by = c("start_station" = "sna")
  )

# 3. 讀取氣象站清單
weather_sta <- read.csv(
  "Data/weather_sta_list_2025-06-09.csv",
  fileEncoding = "UTF-8"
)

weather_north <- weather_sta %>%
  filter(
    城市 %in% c("臺北市", "新北市"),
    !is.na(經度),
    !is.na(緯度),
    撤站日期 == ""
  )

weather_sf <- st_as_sf(
  weather_north,

```

```

  coords = c("經度", "緯度"),
  crs = 4326
) %>%
  st_transform(3826)
# 讀入已整理好的 weather panel

rain_panel <- readRDS("Output/rain_panel.rds")

temp_panel <- readRDS("Output/temp_panel.rds")

humidity_panel <- readRDS("Output/RelativeHumidity.rds")

# 1. rain 最近測站
rain_station_ids <- unique(rain_panel$weather_station_id)
weather_rain_sf <- weather_sf %>%
  filter(站號 %in% rain_station_ids)
nearest_rain_idx <- st_nearest_feature(
  bike_sf,
  weather_rain_sf
)
bike_sf$nearest_rain_station <- weather_rain_sf$站名
[nearest_rain_idx]
bike_sf$nearest_rain_id <- weather_rain_sf$站號
[nearest_rain_idx]
# 2. temperature 最近測站
temp_station_ids <- unique(temp_panel$weather_station_id)
weather_temp_sf <- weather_sf %>%
  filter(站號 %in% temp_station_ids)
nearest_temp_idx <- st_nearest_feature(
  bike_sf,
  weather_temp_sf
)
bike_sf$nearest_temp_station <- weather_temp_sf$站名
[nearest_temp_idx]
bike_sf$nearest_temp_id <- weather_temp_sf$站號
[nearest_temp_idx]

# 3. humidity 最近測站

```



```

humidity_station_ids <-
unique(humidity_panel$weather_station_id)
weather_humidity_sf <- weather_sf %>%
  filter(站號 %in% humidity_station_ids)
nearest_humidity_idx <- st_nearest_feature(
  bike_sf,
  weather_humidity_sf
)

bike_sf$nearest_humidity_station <- weather_humidity_sf$站
名[nearest_humidity_idx]

bike_sf$nearest_humidity_id <- weather_humidity_sf$站號
[nearest_humidity_idx]

# 建立 YouBike station ↔ weather station 對照表

bike_weather_map <- bike_sf %>%
  st_drop_geometry() %>%
  mutate(
    sna = gsub("YouBike2.0_", "", sna)
  ) %>%
  select(
    sno,
    sna,
    sarea,
    nearest_rain_station,
    nearest_rain_id,
    nearest_temp_station,
    nearest_temp_id,
    nearest_humidity_station,
    nearest_humidity_id
  )

mrt_sub_Youbike_all <- mrt_sub_Youbike_all %>%
  left_join(
    bike_weather_map %>%
      select(

```

```

      sna,
      nearest_rain_station,
      nearest_rain_id,
      nearest_temp_station,
      nearest_temp_id,
      nearest_humidity_station,
      nearest_humidity_id
    ),
    by = c(
      "start_station" = "sna"
    )
  )
)

mrt_hourly <- mrt_sub_Youbike_all %>%
  filter(
    hour %in% c(6:23)
  ) %>%
  mutate(weekday = weekdays(date), weekend =
ifelse(weekday %in% c("Saturday", "Sunday"), 1, 0)
  ) %>%
  group_by(
    mrt_station,
    date,
    hour
  ) %>%
  summarise(
    trip_count = n(),
    weekday = first(weekday),
    weekend = first(weekend),
    nearest_rain_station = first(nearest_rain_station),
    nearest_rain_id = first(nearest_rain_id),
    nearest_temp_station = first(nearest_temp_station),
    nearest_temp_id = first(nearest_temp_id),
    nearest_humidity_station =
first(nearest_humidity_station),
    nearest_humidity_id = first(nearest_humidity_id),
    .groups = "drop"
  )

```

```

mrt_weather <- mrt_hourly %>%
  left_join(
    rain_panel,
    by = c("nearest_rain_id" = "weather_station_id",
"date", "hour"))

# merge temperature

mrt_weather <- mrt_weather %>%
  left_join(
    temp_panel,
    by = c(
      "nearest_temp_id" =
        "weather_station_id",
      "date",
      "hour"
    )
  )

# merge humidity

mrt_weather <- mrt_weather %>%
  left_join(
    humidity_panel,
    by = c(
      "nearest_humidity_id" =
        "weather_station_id",
      "date",
      "hour"
    )
  )

mrt_weather_clean <- mrt_weather %>% drop_na()

#建立一個 rain_dummy
mrt_weather_clean <- mrt_weather_clean %>%

```

```

mutate(rain_dummy = ifelse(rain > 0, 1, 0))

#borrow
borrow_data <- Youbike_all %>%
  mutate(
    near_mrt = start_station %in%
mrt_bike_station_map$sna,
    hour = hour(start_time)
  ) %>%
  filter(
    near_mrt,
    start_station != end_station
  )
return_data <- Youbike_all %>%
  mutate(
    near_mrt = end_station %in% mrt_bike_station_map$sna,
    hour = hour(start_time)
  ) %>%
  filter(
    near_mrt,
    start_station != end_station
  )
borrow_hourly <- borrow_data %>%
  count(hour) %>%
  rename(count = n) %>%
  mutate(type = "Borrow")
return_hourly <- return_data %>%
  count(hour) %>%
  rename(count = n) %>%
  mutate(type = "Return")
borrow_return_hourly <- bind_rows(
  borrow_hourly,
  return_hourly
)

saveRDS(mrt_weather_clean,
"Output/mrt_weather_clean.rds")

```

```
saveRDS(borrow_return_hourly,  
"Output/borrow_return_hourly.rds")
```

C.3 : EDA

```
library(ggplot2)  
library(dplyr)  
library(tidyr)  
library(scales)  
library(gt)
```

```
mrt_weather_clean <-  
readRDS("Output/mrt_weather_clean.rds")  
#1. trip_count  
mrt_weather_clean %>% ggplot(aes(trip_count)) +  
geom_histogram()  
mrt_weather_clean %>% ggplot(aes(x = log(trip_count+1)))  
+ geom_histogram()
```

不同站點平均需求

```
station_mean <- mrt_weather_clean %>%  
  group_by(mrt_station) %>%  
  summarise(  
    avg_trip = mean(trip_count, na.rm = TRUE),  
    sd_trip = sd(trip_count, na.rm = TRUE),  
    n = n()  
  ) %>%  
  arrange(desc(avg_trip))
```

```
f1 <-  
  station_mean %>%  
  slice_head(n = 20) %>%  
  ggplot(  
    aes(  
      x = reorder(mrt_station, avg_trip),  
      y = avg_trip  
    )  
  ) +  
  geom_col() +
```

```

coord_flip() +
labs(
  title = "Top 20 MRT Stations by Average YouBike
Demand",
  x = "MRT Station",
  y = "Average Trip Count"
) +
theme_minimal(base_family = "PingFang TC")

ggsave(
  "Figures/figure1.png",
  plot = f1,
  width = 8,
  height = 5,
  dpi = 300
)

# YouBike 使用量的日內變化
hourly_pattern <- mrt_weather_clean %>%
  group_by(hour) %>%
  summarise(
    avg_trip = mean(trip_count, na.rm = TRUE)
  )

f2 <-
  hourly_pattern %>% ggplot(aes(x = hour, y = avg_trip))+
  geom_line(size = 1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = 6:23) +
  labs(
    title = "Hourly Pattern of YouBike Usage",
    x = "Hour of Day",
    y = "Average Trip Count"
  ) + theme_minimal()
ggsave(
  "Figures/figure2.png",
  plot = f2,
  width = 8,

```

```

    height = 5,
    dpi = 300
)

# 假日與平日的變化
hourly_weekend <- mrt_weather_clean %>% group_by(hour,
weekend) %>%
  summarise(avg_trip = mean(trip_count), .groups = 'drop')

f3 <-
  hourly_weekend %>%
  ggplot(aes(x = hour, y = avg_trip, color =
factor(weekend))) +
  geom_line(size = 1.2) +
  scale_x_continuous(
    breaks = 0:23
  ) +
  labs(
    title = "Hourly YouBike Demand",
    x = "Hour",
    y = "Average Trip Count",
    color = "Weekend"
  ) + theme_minimal()
ggsave(
  "Figures/figure3.png",
  plot = f3,
  width = 8,
  height = 5,
  dpi = 300
)

# 雨量
rain_hour <- mrt_weather_clean %>%
  group_by(hour, rain_dummy) %>%
  summarise(
    avg_trip = mean(trip_count, na.rm = TRUE),
    .groups = "drop"
  )

```

```

rain_hour <- rain_hour %>%
  mutate(
    rain_label = ifelse(rain_dummy == 1, "Rain", "No
Rain")
  )

f4 <- rain_hour %>%
  ggplot(aes(x = hour, y = avg_trip, color = rain_label,
group = rain_label)) +
  geom_line(linewidth = 1.2) +
  geom_point() +
  scale_x_continuous(breaks = 0:23) +
  labs(
    x = "Hour",
    y = "Average Trip Count",
    color = "Rain Condition",
    title = "Hourly YouBike Demand by Rain Condition"
  ) +
  theme_minimal() +
  theme(
    legend.position = "right"
  )
ggsave(
  "Figures/figure4.png",
  plot = f4,
  width = 8,
  height = 5,
  dpi = 300
)

#温度
temp_pattern <- mrt_weather_clean %>%
  group_by(temperature) %>%
  summarise(avg_trip = mean(trip_count), .groups = "drop")

f5 <- temp_pattern %>% ggplot(aes(x = temperature, y =
avg_trip)) +

```



```

    geom_point(alpha = .5) +
    geom_smooth()
ggsave(
  "Figures/figure5.png",
  plot = f5,
  width = 8,
  height = 5,
  dpi = 300
)

#晴天與雨天的溫度差異
temp_rain <- mrt_weather_clean %>%
  group_by(temperature, rain_dummy) %>%
  summarise(
    avg_trip = mean(trip_count),
    .groups = "drop"
  )

f6 <- temp_rain %>%
  ggplot(aes(temperature, avg_trip, color =
factor(rain_dummy, labels = c("No Rain", "Rain"))
  )
  )
) +
  geom_smooth(se = FALSE) +
  labs(
    color = "Weather Condition"
  )

ggsave(
  "Figures/figure6.png",
  plot = f6,
  width = 8,
  height = 5,
  dpi = 300
)

#濕度

```

```

humidity_pattern <- mrt_weather_clean %>%
  group_by(humidity) %>%
  summarise(
    avg_trip = mean(trip_count)
  )

f7 <- humidity_pattern %>%
  ggplot(aes(x = humidity, y = avg_trip)) +
  geom_point(
    alpha = 0.5,
    size = 2
  ) +
  geom_smooth(
    method = "loess",
    se = TRUE,
    linewidth = 1.2
  ) +
  labs(
    title = "Relationship Between Humidity and YouBike
Demand",
    x = "Humidity (%)",
    y = "Average Trip Count"
  ) +
  theme_minimal(base_family = "PingFang TC") +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
ggsave(
  "Figures/figure7.png",
  plot = f7,
  width = 8,
  height = 5,
  dpi = 300
)

```

```

#
station_rain <- mrt_weather_clean %>%
  group_by(mrt_station, rain_dummy) %>%
  summarise(
    avg_trip = mean(trip_count),
    .groups = "drop"
  ) %>%
  pivot_wider(
    names_from = rain_dummy,
    values_from = avg_trip
  ) %>%
  mutate(
    rain_effect = (`1` - `0`) / `0`
  )

head(station_rain[order(station_rain$rain_effect),], 20)

t1 <- station_rain %>%
  arrange(rain_effect) %>%
  slice_head(n = 20) %>%
  gt() %>%
  cols_label(
    mrt_station = "MRT Station",
    `0` = "Mean Demand (No Rain)",
    `1` = "Mean Demand (Rain)",
    rain_effect = "Percentage Change"
  ) %>%
  fmt_percent(
    columns = rain_effect,
    decimals = 1
  )

gtsave(
  t1,
  "Tables/table1.png"
)

#borrow and return

```

```

readRDS("Output/borrow_return_hourly.rds")
library(ggplot2)

f10 <- ggplot(
  borrow_return_hourly,
  aes(
    x = hour,
    y = count,
    color = type
  )
) +
  geom_line(size = 1) +
  geom_point() +
  scale_x_continuous(
    breaks = 0:23
  ) +
  labs(
    title = "Borrow vs Return Demand by Hour",
    x = "Hour",
    y = "Number of Trips",
    color = ""
  ) +
  theme_minimal()
ggsave(
  "Figures/figure10.png",
  plot = f10,
  width = 8,
  height = 5,
  dpi = 300
)

# 騎乘距離比例
distance_data <- mrt_sub_Youbike_all %>%
  filter(!is.na(ride_distance_km))
f9 <- ggplot(
  distance_data,
  aes(x = ride_distance_km)
) +

```

```

geom_density(
  fill = "#4E79A7",
  alpha = .5
) +
geom_vline(
  xintercept = 0.726,
  color = "red",
  linetype = 2
) +
geom_vline(
  xintercept = 1.872,
  color = "darkgreen",
  linetype = 2
) +
coord_cartesian(xlim = c(0, 5)) +
labs(
  title = "Distribution of MRT-related YouBike Riding
Distance",
  subtitle = "Median = 0.73 km, 90th percentile = 1.87
km",
  x = "Riding Distance (km)",
  y = "Density"
) +
theme_minimal(base_size = 14)
ggsave(
  "Figures/figure9.png",
  plot = f9,
  width = 8,
  height = 5,
  dpi = 300
)

```

C.4 : Model

```

library(dplyr)
library(modelsummary)
library(performance)
library(broom)

```

```

library(ggplot2)
mrt_weather_clean <-
readRDS("Output/mrt_weather_clean.rds")

#檢查相關問題
mrt_weather_clean %>%
  select(rain_dummy,
         temperature,
         humidity,
         weekend) %>%
  cor(use = "complete.obs")

# Model0: Weather only
m0 <- lm(
  trip_count ~
    rain_dummy +
    temperature +
    I(temperature^2) +
    humidity,
  data = mrt_weather_clean
)
summary(m0)

# Model 1: Weather + Time
m1 <- lm(
  trip_count ~
    rain_dummy +
    temperature +
    I(temperature^2) +
    humidity +
    weekend +
    factor(hour),
  data = mrt_weather_clean
)
summary(m1)

# Model 2: Weekend × Hour

```

#檢查平日與假日的日內需求型態是否不同

```
m2 <- lm(
  trip_count ~
    rain_dummy +
    temperature +
    I(temperature^2) +
    humidity +
    weekend * factor(hour),
  data = mrt_weather_clean
)
summary(m2)
# Model 3: Rain × Hour
# 檢查雨天效果是否因時段不同而異
```

```
m3 <- lm(
  trip_count ~
    rain_dummy * factor(hour) +
    temperature +
    I(temperature^2) +
    humidity +
    weekend,
  data = mrt_weather_clean
)
summary(m3)
```

Model 4: Full model with MRT station fixed effects
#控制不同捷運站本身的需求差異

```
m4 <- lm(
  trip_count ~
    rain_dummy +
    temperature +
    I(temperature^2) +
    humidity +
    weekend +
    factor(hour) +
    factor(mrt_station),
```

```

    data = mrt_weather_clean
  )
summary(m4)

# Model 5: Rain × Weekend
m5 <- lm(
  trip_count ~
    rain_dummy * weekend +
    temperature +
    I(temperature^2) +
    humidity +
    factor(hour) +
    factor(mrt_station),
  data = mrt_weather_clean
)
summary(m5)

# Model performance comparison
model_performance <- compare_performance(
  m0, m1, m2, m3, m4, m5
)
model_performance

# 4. Export model comparison table
modelssummary(
  list(
    "M0" = m0,
    "M1" = m1,
    "M2" = m2,
    "M3" = m3,
    "M4" = m4,
    "M5" = m5
  ),
  coef_omit = "factor\\(hour\\)|factor\\(mrt_station\\)",
  output = "Tables/regression.png"
)

```



```

#
station_fe <- tidy(m4) %>%
  filter(grepl("factor\\(mrt_station\\)", term)) %>%
  mutate(
    station = gsub("factor\\(mrt_station\\)", "", term)
  ) %>%
  select(station, estimate)

anova(m0, m1)
anova(m1, m2)
anova(m1, m3)
anova(m1, m4)

# Top 10
top10 <- station_fe %>%
  arrange(desc(estimate)) %>%
  slice(1:10)

# Bottom 10
bottom10 <- station_fe %>%
  arrange(estimate) %>%
  slice(1:10)

plot_station <- bind_rows(
  top10,
  bottom10
)

f8 <- ggplot(plot_station,
  aes(x = reorder(station, estimate),
    y = estimate)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top and Bottom MRT Station Fixed Effects",
    subtitle = "Controlling for weather and time factors",
    x = NULL,
    y = "Station Fixed Effect"
  )

```

```

) +
  theme_minimal(base_family = "PingFang TC")
ggsave("Figures/figure8.png", f8, width = 8, height = 5,
dpi = 300)

```

C.4 : Appendix

```

## nb model
library(MASS)
mrt_weather_clean <-
readRDS("Output/mrt_weather_clean.rds")

model_nb <- glm.nb(
  trip_count ~ rain_dummy + temperature + I(temperature^2)
+
  humidity + weekend + factor(hour) +
factor(mrt_station),
  data = mrt_weather_clean
)
summary(model_nb)
modelssummary(
  list(
    "OLS" = m4,
    "Negative Binomial" = model_nb
  ),
  coef_omit = "factor\\(",
  coef_map = c(
    rain_dummy = "Rain Dummy",
    temperature = "Temperature",
    "I(temperature^2)" = "Temperature2",
    humidity = "Humidity",
    weekend = "Weekend"
  ),
  estimate = "{estimate}{stars}",
  statistic = "({std.error})",
  stars = c('*' = .05, '**' = .01, '***' = .001),
  output = "Appendix_Robustness.png"
)

```